

Designing Dissolving Wearables

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Figure 1: Examples of dissolving wearables. a) “Seasonal Footwear” (crochet) loses the straps after dissolving, becoming more open to suit warmer seasons. b) “Unfolding Lace Top” (knit) unfolds into a new design upon dissolving.

ABSTRACT

Bio-based materials facilitate the development of more sustainable devices and wearables, expanding the range of design possibilities beyond conventional materials. Our work with *biofoam* explores one such quality, *dissolving*, as a unique affordance for designing and interacting with wearables. We developed techniques to make biofoam yarns, and used them to craft three wearables: “Seasonal Footwear”, a “Reveal Bralette”, and an “Unfolding Lace Top”. These wearables incorporate sections that dissolve in water, allowing customization to suit the user’s needs. These wearables illustrate short-term use cases, such as a one-time reveal or shape change. We explore this novel design space as sustainable *ephemeral fashion*, where bio-based dissolving materials enable revealing, transformative, and interactive functionalities.

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CCS CONCEPTS

• **Human-centered computing** → *Interactive systems and tools*.

KEYWORDS

bio-based materials, wearables, design to dissolve, textile craft

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1 INTRODUCTION

In recent years, the wearables and HCI communities have increasingly shown interest in incorporating bio-based materials into their works, aiming for more sustainable interactive wearables. Mycelium skin and SCOBY leather have been used as scaffolding for embedding electronics, offering unique interactivity by changing texture and appearance over time [3, 21, 22]. Material traces resulting from these changes can enhance traditional forms of interactivity [17, 19],

while the biodegradable nature of these materials allows for the reharvesting of electronic components at the end of the wearables' life [2, 10]. Besides sustainability benefits, bio-based materials also possess other less explored qualities such as their ability to dissolve in water or be shaped in ways apart from molding and layering [2, 10]. Thus, we believe that expanding the current fabrication techniques could open new design opportunities to integrate bio-based materials in wearables and textiles design. This paper further explores biofoam [10], a gelatin-based foam with the unique quality of *dissolving*, making it an intentional design affordance for short-term fashion wearables. We present an expanded fabrication process for biofoam strings, resembling yarns, and demonstrate the creation of three dissolving wearables through weaving, crocheting, and knitting techniques. These wearables offer sustainable fashion possibilities. Additionally, we discuss design considerations and opportunities that arise from using traditional textile craft techniques and their limitations when designing dissolving wearables.

2 RELATED WORK

Our work on dissolving bio-based materials aligns with the growing trend of sustainable designs and artistic explorations. In the realm of sustainable fashion, bio-based materials offer innovative alternatives to conventional fashion materials. Fruit Rotterdam, for instance, utilizes local fruit waste streams to create leather-like fabrics [18]. Furthermore, designer Suzanne Lee crafted a jacket made out of kombucha SCOBY (Symbiotic Culture Of Bacteria and Yeast), and Aniela Hoitink from Dutch start-up NEFFA [14] made pieces of clothing from mycelium. The DIYBio [6] and wearable technology communities in HCI have also embraced bio-based materials for creating interactive wearables [24], displays [1], or growable devices [9, 22], or to design embodied interactions [15, 16]. While many of these designs rely on molding and layering fabrication techniques [21, 22], hand-craft techniques have been rarely employed. Regarding dissolving textiles, textile artists and craftspeople have been designing with dissolving materials for years. For instance, they have used commercially-available dissolving thread to temporarily secure quilt layers, albeit not exclusively using bio-based materials. Chalayan's Spring/Summer 2016 collection featured dresses that dissolved to reveal Swarovski-crystal-embellished gowns [11]. Textile artist Sasha de Koninck used dissolving embroidery materials in her "Knotting.Knotted.Knot" work [5], washing pieces away to explore states of emotion and anxiety, and designer Sofia Guridi integrated biomaterials that dissolve to reveal hidden images in their textile piece "Cambio de Piel - Change of skin" [8]. We believe that dissolving bio-based materials can be a compelling feature for both function and aesthetics, opening new possibilities for expression.

In previous work, designers acknowledge that working with bio-based materials introduces specific values into the design process. For instance, working with grown materials requires attention to slowness [15] and intentionality [3] due to the slower growth of living organisms compared to manufactured outputs. Similarly, employing materials from food waste demands innovative fabrication techniques to shape bio-based raw materials into various forms, bringing design considerations of imperfection and uncertainty [4]. Expanding on this line of research, we explore the dissolving quality of *biofoam* through three textile applications and envision its

integration into daily life scenarios. Our focus lies on yarn-based textile craft techniques, reflecting on the process of working with bio-based materials, both before and after dissolving.

3 DISSOLVING WEARABLES

In this study, we explore *dissolving* as an intentional design affordance for fashion wearables. We demonstrate this concept through three use cases, showcasing the potential of biofoam yarns with dissolving attributes when combined with other materials using traditional textile craft techniques such as weaving, crocheting, and knitting. The selection of a bio-based material was guided by considerations of comfort and tactile experience. Moreover, this work serves as an initial effort to gain insights into the behavior of biofoam yarns when blended with other materials and subjected to various levels of stress during fabrication and wearing.

3.1 Fabricating Biofoam Strings and Yarns

Biofoam strings, resembling yarns, serve as the primary "building blocks" for crafting "Seasonal Footwear", "Reveal Bralette", and an "Unfolding Lace Top". While the detailed process to create biofoam is documented in our previously published paper with the open-source recipe [10], for this study, we used recycled biofoam scraps available in our lab. Moreover, we developed a tutorial¹ presented at a DIS'23 workshop [20], outlining the fabrication of biofoam strings, which can be found as supplemental material. To obtain the strings, we devised a user-friendly technique adapted from traditional cake piping. Employing syringes of varying diameters and pipe bags, we meticulously extruded the biofoam into strings of diverse thicknesses and textures. Below, we outline the steps involved in our process.

Melting biofoam scraps: We chopped the biofoam scraps into small pieces using scissors. Then, we added half a cup of chopped biofoam to 20 ml of boiling water in a pot. Keeping the mixture at a constant temperature of 140°F, we stirred with a silicone spatula until the biofoam scraps fully melted. One of the challenges at this phase occurred when there were density inconsistencies (lumps) in the biofoam scraps. In this case, the gelatin, the main component of the biofoam, visually started sticking to the bottom of the pot. To address this, we added the same initial amount of water to the pot (20 ml), removed the pot from the heat, and continued stirring with the silicone spatula. After the mixture became more homogeneous, we placed it back on the stove at 140°F and stirred until reaching a "soft" or "stiff" peaks meringue stage (Fig.2b, 2c).

Making biofoam meringue: In successful biofoam extrusion, the meringue stage (foamy, soft, stiff, or broken) plays a crucial role. Paying close attention to this step is essential. If the biofoam meringue is in its *foamy stage*, the strings will contain numerous air bubbles that evaporate during drying, resulting in a flattened shape (Fig.2a,e). An increased amount of air also affects the follow-up interaction, as airy strings tend to compress upon touch. Piping the biofoam at the *soft peak stage* creates a half-moon cross-section in the strings (Fig.2b,e). This stage is suitable for novice crafters as it allows time for testing extrusion steadiness and allows for trial and error. Piping at the *stiff peak stage* produces the most rounded strings (Fig.2c,e). This stage is ideal for experienced crafters who

¹How to make biofoam strings. <https://www.softrobotics.io/post/how-to-make-soft-biofoam-strings-exploring-an-affordable-and-accessible-fabrication-method>

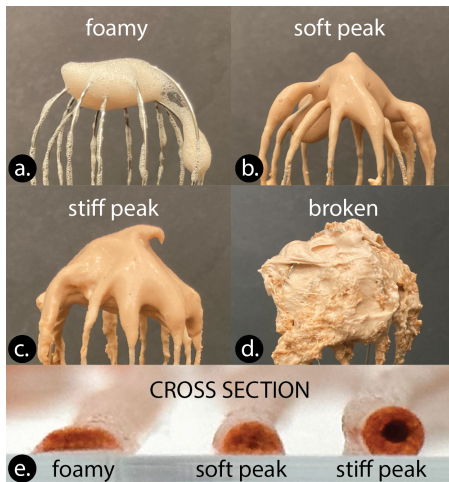


Figure 2: Different consistencies of biofoam meringue: a. foamy, b. soft peak, c. stiff peak, d. broken, e. cross-section of biofoam yarns at different consistencies.

can maintain a constant extrusion speed to avoid clogging. The mixture hardens quickly at this stage, requiring fast and steady extrusion. If the biofoam meringue is in a *broken stage* (Fig.2d), typically due to excessive stirring, it is best to fix the consistency or start over. Extruding from a piping bag with a broken meringue is likely to be unsuccessful due to the high risk of clogging. To fix the consistency, we recommend adding a bit of hot water and stirring until soft peaks appear on the spatula.

Extruding biofoam strings and making yarns: We used a pipe decorating tip set² to extrude biofoam strings. The extruded strings were dried for at least 12 hours on a non-stick surface, like acrylic or non-stick silicone, to ensure proper curing and reduce stickiness. To prevent clogging during extrusion, we kept a cup of hot water nearby and dipped the metal tip in it for one minute when necessary (see supplemental material for illustration). Initially, our extrusion process was slow, leading to quick setting inside the bag. In such cases, we had to put the biofoam back in the pot and recook it to restart the process of making biofoam meringue. After extensive experimentation, we found that using only piping tips with 1 and 2-mm diameters was the best option for producing larger amounts of biofoam strings while preserving the original color of the biofoam scraps. The volume of our piping bags allowed a maximum length of 1 m for the extruded strings. To elongate the strings, we developed two methods to adhere multiple strings together. The first method involved dipping one end of a string in warm water and immediately placing it on top of a second dry biofoam string end. Water caused the biofoam string to swell and function as glue. We then pressed the two string ends together for a couple of seconds and released the pressure afterward to verify successful joining. The second method, spinning, was used only when making 2-ply biofoam yarns.

3.2 Use Cases

We used biofoam yarns to craft dissolving wearable fashions, envisioning three use cases for this feature: 1) “Seasonal Footwear”,

2) “Reveal Bralette”, and 3) “Unfolding Lace Top”. In the first two cases, users intentionally trigger the dissolution process, while the third case explores how weather conditions like rain or humidity might contribute to the wearable’s dissolution or shape change. As the biofoam yarns dissolve, they leave behind non-dissolving yarns, forming new wearable patterns and structures.

Seasonal Footwear: We crafted this wearable, as shown in Figure 1a, by customizing a crocheted sandal that blends biofoam, cotton yarn, and a flip-flop base. After wearing them in the Spring, the sandals can be transformed into more open sandals for the Summer. The user triggers the transformation intentionally by washing the sandals, causing the biofoam to dissolve. In Figure 1b, the Summer sandal results from one machine cycle with warm water. As Spring sandals, they feature heel and ankle support entirely crocheted with biofoam yarn, which dissolves completely during washing. The top of the vamp is crocheted with a blend of cotton and biofoam yarn, partially dissolving as the biofoam yarn serves a decorative purpose without compromising the wearing comfort and stretchiness of the sandals. We made several iterations, deciding which sections to create with biofoam and which with standard cotton. To cover the rubber strap, we applied a single crochet stitch, creating a base for continuous crocheting. Using a blend of cotton and biofoam yarn, we employed a single crochet stitch to cover the top section of the sandal, reducing stitches to accommodate the triangle shape. We crafted a back strap out of biofoam to secure the Achilles heel and upper ankle, attaching it to the side straps. Finally, we created a lace by chaining and threaded it through the top ankle strap.

Reveal Bralette: We crocheted a bralette that through *dissolving* reveals a new design upon interacting with water (Figure 3). The bralette features light blue biofoam yarn, matching the color of other yarns and concealing the dissolving elements. The bralette’s overall design draws from various crocheted granny square designs, traditional motifs that are also popular in contemporary crochet fashion. We selected a granny square design with long, spoke-like stitches (dark blue) to ensure structural integrity after the biofoam yarn dissolved. Alternating rings of biofoam and cotton yarns were supported by these non-dissolving stitches, and we secured them with matching yarn to prevent unraveling when the biofoam dissolved. Our approach not only prevents the garment from “falling apart” but also allows us to intentionally create a design that changes upon dissolution. For example, we designed the bottom border to transform, hanging differently after interacting



Figure 3: a. Crocheted bralette chest before dissolving. b. After dissolving the light blue biofoam, patches become hollow and reveal the color underneath.

²Wilton 55-Piece Cake Decorating Tip Set. <https://tinyurl.com/4bzvcdum>

with water (Figure 3b). By using chain stitches made with either biofoam or light-blue yarn, specific stitches dissolve upon contact with water, releasing the orange yarn strings and bringing a new dimension to the design. We envision wearing the *Reveal Bralette* at a festival where water exposure could trigger a transformation, presenting a color pop or transitioning to a more lacey pattern (Figure 3b). This concept could be extended to full clothing pieces, where a second skin is revealed as the soluble material interacts with warm water, offering novel possibilities in fashion design.

Unfolding Lace Top: We designed this wearable to use *dissolving* for altering its shape (Figure 1b). Combining hand-knitting techniques, we created an unfolding stitch structure and knitted a sample swatch to test different stitches and yarns (Fig. 4). The structure comprises densely knitted cords with lace bands in between. The lace portions use thinner yarn and open stitches, forming a lighter-weight fabric. The biofoam yarn is incorporated between cords, scrunching the lace into a pleat. To construct the garment, we knitted a bra as the base and sewed in the lace panel. By applying water using a spray bottle while wearing the top, we dissolved the biofoam gradually. Some segments of the garment unfolded first, while others remained folded, depending on the water application. Agitation from the wearer’s movements accelerated the dissolution. We hypothesize that unfolding may be influenced by how water contacts the garment, whether intentionally applied or due to environmental factors like rain. The “Unfolding Lace Top” structure offers potential applications in various pieces throughout a garment, such as a jumpsuit, resulting in more striking changes in shape and silhouette. Even with multiple identical garments made from a single design, each one would undergo a distinct transformation based on water exposure, including weather conditions and wearer movements. For instance, we envision creating a set of dance costumes for a chance-based performance series, where each dance yields a unique shape.

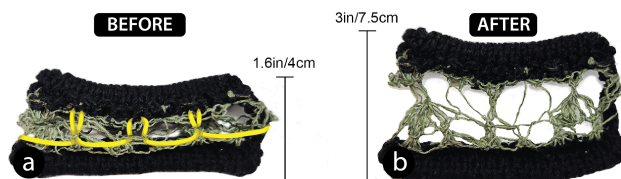


Figure 4: “Unfolding Lace Top” swatch to test the unfolding stitch structure. a. before, and b. after dissolving the swatch in cold water.

4 DISCUSSION AND CONCLUSION

While our wearables did not incorporate electronics in their design, we believe our fabrication process and hand-craft techniques could support future prototypes that integrate electronics alongside the dissolving materials to make interactive wearables. Dissolving bio-based materials in wearable electronic textiles could support the current body of research within HCI of design for decay and decompose [3, 12, 22], disassembly [23], and unmaking [13]. All of which can be tied to sustainability goals. With this understanding, we could develop more sustainable e-textiles using “dissolving” as an interactive feature. For instance, when a moisture sensor is

embedded in a knitted garment that incorporates biofoam yarn, the contact of the garment with sweat or moisture (e.g., from the wearer’s skin) could initiate the biofoam yarn dissolution, thereby triggering the sensor to measure and transmit data. Conversely, the garment could contain embedded electronics that actively control the dissolving process, utilizing mechanisms such as small water capsules that activate when compressed, thereby enabling the development of programmable dissolvable garment designs. Our paper additionally facilitates the reduction of barriers to entry for hand-craft methods in prototyping dissolving wearables since current innovations and research in sustainable wearable electronic textiles require specialized equipment to make and use their materials [7]. Furthermore, we would like our fabrication method could be used to inform future tool designs in wearable technology research.

Reflecting on textile craft techniques: We ended up choosing techniques in which we can customize the structure or stitch density of the garment. Thus, knitting and crocheting showed to be the techniques that support such customization over weaving. For instance, while the dissolving or unraveling of woven structures leads to full garment disassembly, knitting or crocheting allowed us to explore other design attributes such as revealing or unfolding. Furthermore, crocheting with blended biofoam and cotton yarn, allowed us to form a tight mesh, that after dissolving the mesh freed up and became airier showing a change in stitch density (Fig. 1b, 4).

Designing to dissolve: When designing dissolving wearables, we adopted a body-conscious approach, carefully considering the placement of biofoam dissolving sections on the body and how the body interacts with the material. This required a different mindset, where designers must detach themselves from their labor, knowing that it will eventually dissolve away. Wearables made with dissolving biofoam materials become ephemeral, with a limited lifespan due to their ability to dissolve upon interaction with water, such as in rain or sweat. However, this ephemeral nature also presents an opportunity for these wearables to record and physicalize aspects like weather or emotions.

Limitations: The biofoam yarn’s characteristics are highly dependent on the environment. For instance, while knitting the *Unfolding Lace Top*, co-author Shanel, traveled from a humid to a drier climate, causing the biofoam yarn to become less sticky and less elastic as it dried out. This difference in tactile experience did not make it easier or more difficult to work with the biofoam, but it was noticeable. Another limitation arises from the residues left behind after the biofoam dissolves, particularly in unintended scenarios like exposure to rain or during water fights at festivals. This also affects the non-water soluble yarn left behind, making it slightly crunchy. This observation led us to recognize the importance of intentionally placing the biofoam, transforming the act into a deliberate design decision rather than a mere experiment.

Conclusion: Our paper explores *dissolving* as a design affordance that offers novel design possibilities for sustainable and interactive wearables. Our research showcases three dissolving wearables crafted with water-soluble biofoam yarns, highlighting short-term applications such as one-time reveals and shape changes. The fabrication process and hand-craft techniques we used offer a potential pathway for developing future prototypes that combine electronics with dissolving materials, enabling the creation of ephemeral interactive wearables.

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